

2012-2013 – Master 2 – Macro I

Lecture 4 : The Dixit-Stiglitz preference structure

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(based on Gilles Saint-Paul lecture notes)

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Changes from version 1.0 are in red

Changes from version 1.1 are in purple

Disclaimer

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0. Introduction

- ▶ The Dixit-Stiglitz apparatus is a central building block of modern macroeconomic theory, in both growth and business cycles. Its interest is three-fold :
 - ▶ It delivers simple demand functions for a continuum of atomistic producers, which is useful for analysing price setting and monopolistic competition
 - ▶ The number of goods can be endogenized, which is useful for growth and R&D
 - ▶ It is easy to aggregate things over a sector or the whole economy, which is useful for moving from micro relationships to macro ones.

1. The Model

Households problem

- ▶ A representative consumer has preferences $u(C)$, where C is a CES aggregate consumption index given by

$$C = \left(\int_0^N c_i^\alpha di \right)^{1/\alpha}. \quad (1)$$

- ▶ Thus there is a continuum of goods indexed by i and consumption of good i is c_i . The total mass of goods is N . Let p_i be the price of good i and Y the income of the consumer. He maximizes C or equivalently C^α :

$$\int_0^N c_i^\alpha di$$

subject to

$$\int_0^N p_i c_i di \leq Y.$$

1. The Model

Households FOC

- ▶ The Lagrangian is

$$\mathcal{L} = \int_0^N c_i^\alpha di + \lambda \left(Y - \int_0^N p_i c_i di \right).$$

- ▶ The FOC is

$$\alpha c_i^{\alpha-1} = \lambda p_i.$$

- ▶ We can write this as

$$c_i = \frac{\alpha^\sigma}{\lambda^\sigma p_i^\sigma}, \quad \text{where } \sigma = \frac{1}{1-\alpha}. \quad (2)$$

- ▶ **Remark :** $\alpha \leq 1$, otherwise the utility function would not be concave.
- ▶ We get the value of λ by substituting (2) into the budget constraint :

$$\lambda = \alpha \left(\frac{Y}{\int_0^N p_i^{1-\sigma} di} \right)^{-\frac{1}{\sigma}}.$$

1. The Model

Households FOC (continued)

- ▶ Hence :

$$c_i = \frac{Y}{p} \left(\frac{p_i}{p} \right)^{-\sigma},$$

- ▶ p is the aggregate price index given by

$$p = \left(\int_0^N p_i^{1-\sigma} di \right)^{\frac{1}{1-\sigma}}.$$

- ▶ Note that

$$\begin{aligned} C &= \left(\int_0^N c_i^\alpha di \right)^{1/\alpha} = \left(\int_0^N c_i^{\frac{\sigma-1}{\sigma}} di \right)^{\frac{\sigma}{\sigma-1}} \\ &= \frac{Y}{p} \left(\int_0^N \left(\frac{p_i}{p} \right)^{1-\sigma} di \right)^{\frac{\sigma}{\sigma-1}} \\ &= \frac{Y}{p}. \end{aligned}$$

2. The role of product variety

The role of σ

- ▶ Suppose we introduce a small mass dN of additional goods at price $\tilde{p}$. What is the effect on utility?
- ▶ Since we know that $C = Y/p$, we just have to look at the effect on p .
- ▶ We get

$$dp = \frac{1}{1-\sigma} \left(\int_0^N p_i^{1-\sigma} di \right)^{\frac{\sigma}{1-\sigma}} \tilde{p}^{1-\sigma} dN.$$

- ▶ This is < 0 iff $\sigma > 1$.
- ▶ Thus people are made better-off by variety, regardless of the price of new goods, provided goods are not too complements.

2. The role of product variety

Why the $\sigma < 1$ case is meaningless?

- ▶ If $\sigma < 1$, the formula suggests that increased variety reduces utility.
- ▶ This makes no sense since I can always abstain from consuming the new goods and thus replicate the initial allocation.
- ▶ By revealed preferences I cannot be made worse-off.
- ▶ What is going on is that if $\sigma < 1$ the preceding utility function is not nested in the new one when the number of goods goes up.
- ▶ To see this, assume that we have $c_i = 0$ for $N_0 < i \leq N$ and $c_i > 0$ otherwise. For $\sigma > 1$ we have $\alpha > 0$ and therefore

$$\begin{aligned} C &= \left(\int_0^{N_0} c_i^\alpha di + \int_{N_0}^N 0 \cdot di \right)^{1/\alpha} \\ &= \left(\int_0^{N_0} c_i^\alpha di \right)^{1/\alpha}. \end{aligned}$$

2. The role of product variety

Why the $\sigma < 1$ case is meaningless?

- ▶ Thus, utility is the same as if only goods between 0 and N_0 existed. On the other hand, if $\alpha < 0$ we have

$$\begin{aligned} C &= \left(\int_0^{N_0} c_i^\alpha di + \int_{N_0}^N \infty di \right)^{1/\alpha} \\ &= 0 \\ &< \left(\int_0^{N_0} c_i^\alpha di \right)^{1/\alpha}. \end{aligned}$$

- ▶ Intuitively, goods are so complements that not consuming one of those goods makes me as miserable as if I consumed no good at all. But then to be consistent I should also miss those goods I do not consume when they are not around, which is not true with specification (1).